

# Wei-Hsin Chen

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Design engineer who builds at the intersection of what designers dream up and what developers ship — interactive prototypes that become production code, internal tools that give teams creative leverage, and design systems precise enough to scale. At Roblox, shipped a scroll-driven 3D storytelling experience in Three.js, architected a CMS abstraction layer that lets content teams publish independently, and built an AI workflow system that lets designers co-create production-ready components without writing code. Comfortable owning the whole surface: motion, markup, tooling, and the conversation between all three.

## Areas of Expertise

### Front-End Development

- Design Systems
- Component-Based Architecture
- Performance Optimization
- UX/UI Prototyping
- Accessibility(a11y)
- Web Animation

### Software Development

- Web Technologies
- Rapid Prototyping
- Test-Driven Development (TDD)
- AI Workflow Engineering
- Motion Design
- Localization / i18n Systems

### Leadership & Management

- Agile Project Execution
- Stakeholder Engagement
- Cross-functional Collaboration
- Strategic Planning
- Technical Documentation
- Regulatory Compliance

## Experience

### Roblox Corporation — California — 04/2025 - Present **Senior Web Engineer**

*Leading platform architecture, migration, and AI-augmented tooling across Roblox's corporate web properties serving millions of global users.*

#### Platform Migration & Performance

- **Architected and shipped migration** of 4 corporate web properties (60+ components) from AngularJS to Next.js/React within a 4-month timeline, implementing SSR, SSG, ISR, and a Turborepo monorepo.
- **Improved desktop Core Web Vitals** post-launch: LCP 42% faster (3.99s → 2.32s), INP 57% faster (298ms → 129ms), CLS eliminated (0.11 → 0.00) — all metrics now in Google's "good" range.
- **Designed cms-transformers adapter package** with an abstraction layer between CMS, UI components, and applications — enabling the team to connect different CMS sources to the same component library without UI rewrites, and giving content teams the ability to update pages independently.
- **Architected enterprise GA4 analytics framework** capturing 10+ interaction types across 10+ international markets, including a cross-locale normalization solution enabling unified reporting previously unavailable.

#### AI Workflow System

- **Designed and shipped a repo-native AI workflow system** that compresses design-to-production cycles: 14 always-on Cursor rules, 8 execution playbooks, and scoped AGENTS.md files at three directory levels — designers brief a page, AI selects from a 64-component registry against brand spec, and developers receive near-production code to review and ship rather than build from scratch.
- **Built a 64-component registry** with a confidence-gated selection system and a machine-readable DESIGN.md canonical brand spec — single source of truth replacing per-session context setup across all AI tools (Cursor, Claude Code, Codex).

#### Interactive & Design-Engineering

- **Designed and engineered the Engineer site's Cube AI flagship experience** — scroll-driven 3D storytelling in Three.js across six animated sections (zoom, rotation, auto-rotate), plus a CubeGenerator prototype connecting to Hugging Face via Gradio for real-time AI-generated 3D model creation. Took the project from early design prototype to production.
- **Led Safety Center microsite refresh** — introduced Compact theme page layout and new components supporting dynamic content and resource discovery.
- **Led homepage redesign** with scroll text animation, delivering +43.6% week-over-week traffic growth (236K vs. 165K sessions), Join Now CTR of 13.7–20% against a 10% target, and #1 SERP rankings for "Roblox parental controls" and "Corp Roblox" post-launch.

#### Localization, CMS & Compliance

- **Migrated Brand, Corp, and Careers sites to Payload CMS** and integrated DeepL AI translation pipeline, expanding locale coverage from 16 → 24 markets with full content automation and per-locale locking.

- **Led cross-functional CCPA compliance initiative** for enterprise marketing properties, delivering before the regulatory deadline with zero compliance gaps — coordinating Legal, Regulatory, and engineering teams.

## **Huge (Google Embedded Team) — California 07/2016 - 04/2024 Tech Lead → Senior Web Engineer**

Embedded engineer at Huge, client: Google. Led cross-functional teams delivering high-impact marketing solutions for Google's core products.

### **Chrome Enterprise & ChromeOS | Tech Lead | 07/2023 – 04/2024 · Team: 5 Engineers, 2 QA**

- **Engineered a configurable design system** using design tokens and CSS custom properties, reducing project setup time by 60% across multiple products.
- **Developed a Figma-based Webpack plugin** for automated style guide generation, cutting client review cycles from 5 days to 1 day and doubling development velocity.
- **Built a React-inspired, reusable component engine** in TypeScript/Python, improving interface coherence by 40% and boosting team productivity by 200%.

### **Chromebook | Tech Lead | 07/2019 – 07/2021 · Team: 3 Engineers, 2 QA**

- **Led development of Perks Redemption SPA** with Python microservice and CMS integration, increasing user engagement by 30%.
- **Rebuilt Product News Service** with microservice architecture, enhancing data integrity and streamlining platform updates.
- **Revamped Product Detail Pages**, reducing content team workload by 30% through improved CMS implementation.

### **Chrome | Senior Web Engineer → Tech Lead | 06/2017 – 06/2019 · Team: 2 Engineers, 1 QA**

- **Rearchitected Cross-Platform Product Download system**, increasing site performance by 270%, development efficiency by 40%, and reducing update cycles by 80%.
- **Implemented A/B testing strategies** resulting in a 15% increase in user engagement and 10% improvement in overall performance.

## **Additional Experience & Community Involvement**

### **Gopomo, Inc — Remote 09/2012 – 08/2023 Design & Technology Consultant**

Led UX/UI design and development for Taiwan Fortune 500 companies using agile methodologies, improving project turnaround times and client satisfaction.

### **Eveo, Inc — California 01/2009 – 06/2016 Front-End Developer → Senior Web Engineer**

Spearheaded front-end development for Fortune 500 clients, focusing on interactive web applications and enterprise CMS solutions.

## **Technical Proficiencies**

**Core Technologies:** JavaScript (ES6+) | TypeScript | HTML5 | CSS3/SCSS | Python | React | Next.js | Astro

**AI & Automation:** Cursor Rules / AGENTS.md | Claude Code | DeepL AI | Hugging Face / Gradio | Three.js | Prompt Engineering

**Design Systems:** Design Tokens | Figma & Figma API | Adobe Creative Suite | Style Guide Generation | Payload CMS | ButterCMS

**Component Architecture:** React | Astro | Node.js | TypeScript | Atomic Design | Storybook

**Animation:** GSAP | Anime.js | CSS Animations | Scroll-driven Animation | Performance Optimization

**Testing & Quality:** BackstopJS | Cypress | Jest | Lighthouse | Visual Regression

**Build & DevOps:** Webpack | Turborepo | NPM | Git | GitHub | CI/CD | Google Cloud Platform

**Analytics:** GA4 | GTM | UTM Governance | Cross-locales Tracking | Data Layer Architecture

**Methodologies:** Design Thinking | Agile (Scrum, Kanban) | TDD | SDLC

**Languages:** English (fluent) | Chinese / Mandarin (native)

## **Education & Professional Development**

**Master of Fine Arts, Computer Art: New Media** | Academy of Art University - California | 2006 - 2009

**Bachelor of Business Administration (BBA)** | National Chung Cheng University - Taiwan | 1999 - 2003